

Sara McAllister

Assistant Professor, University of Wisconsin-Madison

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Education

Carnegie Mellon University

PHD IN COMPUTER SCIENCE, ADVISORS: PROF. NATHAN BECKMANN AND PROF. GREG GANGER

DISSERTATION: TOWARD SUSTAINABLE DATACENTERS THROUGH EFFICIENT DATA RETRIEVAL

Pittsburgh, PA

Aug 2019 - Aug 2025

Carnegie Mellon University

MASTERS IN COMPUTER SCIENCE RESEARCH

Pittsburgh, PA

Aug 2019 - May 2022

Harvey Mudd College

B.S. IN COMPUTER SCIENCE, GRADUATED WITH HIGH DISTINCTION

Claremont, CA

Aug. 2015 - May 2019

Honors and Awards

2025 Siebel Scholar, for outstanding academic performance and leadership

2024

2023 Rising Star, in Electrical Engineering & Computer Science (EECS)

2023

NDSEG Exemplary Poster Presentation, in computer and computational sciences at NDSEG fellows conference

2023

SIGCSE Best Paper Award, for CS-JEDI paper

2023

CMU Graduate Student Service Award, for the development of 15-996 CS-JEDI

2022

SOSP Best Paper Award, for Kangaroo paper

2021

NDSEG Graduate Fellowship, DoD sponsored 3-year fellowship

2021

NSF Graduate Research Fellowship (GRFP), NSF sponsored 3-year fellowship

2021

Harvey Mudd Class of '94 Award, for an outstanding CS graduate in coursework, research, and service

2019

Harvey Mudd Computer Science Departmental Honors

2019

Harvey Mudd Clinic Team Award, for outstanding performance on an industry-sponsored team capstone project

2019

CRA Outstanding Undergraduate Researcher Award, Honorable Mention

2019

Employment History

Assistant Professor, University of Wisconsin-Madison – Computer Science Department

Aug 2026 - Present

Visiting Faculty Researcher, Google – Sunnyvale, CA

Sept 2025 - Present

Managers: Dr. David Culler, Dr. Kim Keeton, Dr. Hank Levy

Graduate Research Assistant, Carnegie Mellon University – Pittsburgh, PA

Aug 2019 - Aug 2025

Advisors: Prof. Nathan Beckmann and Prof. Greg Ganger

Research Intern, Microsoft Research – Redmond, WA

Summer 2022

Mentor: Dr. Amar Phanishayee

Researched serving LLMs more efficiently especially under failure

Research Intern, Microsoft Research – Redmond, WA

Summer 2021

Mentor: Dr. Daniel Berger

Researched in-kernel disaggregated CXL memory solutions

Software Engineering Intern, Yelp – San Francisco, CA

Summer 2019

Database Team

Designed, implemented, and rolled out a library to manage MySQL database permissions

Clinic (Capstone) Project, Harvey Mudd College – Claremont, CA

Aug 2018 - May 2019

Sponsored by Pure Storage

Technical lead for team of 4 designing failover mechanisms for NFS VMs running on a two-controller system

Undergraduate Research Assistant, UNC Chapel Hill – Chapel Hill, NC

May 2018 - Aug 2018

Advisor: Prof. Don Porter

Researched theoretical and experimental analysis of write-optimized dictionaries

Software Engineering Intern, Facebook – Menlo Park, CA

Summer 2017

Developer Experience Team

Developed a Python library to restart and repair development servers

Peer-Reviewed Publications

Scaling the IO wall with Declarative IO

Sanjith Athlur*, [Sara McAllister*](#), Theo Gregersen, Timothy Kim, Yiwei Chen, Sarvesh Tandon, Lucy Wang, Daniel S. Berger, Saurabh Kadekodi, Arif Merchant, Benjamin Berg, Nathan Beckmann, Rashmi Vinayak, George Amvrosiadis, Gregory R. Ganger (* = co-first author)

OSDI 2026

Acceptance Rate: 20%

FairyWREN: A Sustainable Cache for Emerging Write-Read-Erase Flash Interfaces

[Sara McAllister](#), Yucong Wang, Benjamin Berg, Daniel S. Berger, George Amvrosiadis, Nathan Beckmann, Gregory R. Ganger

ACM TOS

2025

A Call for Research on Storage Emissions

[Sara McAllister](#), Fiodar Kazhamiaka, Daniel S. Berger, Rodrigo Fonseca, Kali Frost, Aaron Ogus, Maneesh Sah, Ricardo Bianchini, George Amvrosiadis, Nathan Beckmann, Gregory R. Ganger

Energy Informatics Review

2025

FairyWREN: A Sustainable Cache for Emerging Write-Read-Erase Flash Interfaces

[Sara McAllister](#), Yucong Wang, Benjamin Berg, Daniel S. Berger, George Amvrosiadis, Nathan Beckmann, Gregory R. Ganger

OSDI 2024

Acceptance Rate: 18%

A Call for Research on Storage Emissions

[Sara McAllister](#), Fiodar Kazhamiaka, Daniel S. Berger, Rodrigo Fonseca, Kali Frost, Aaron Ogus, Maneesh Sah, Ricardo Bianchini, George Amvrosiadis, Nathan Beckmann, Gregory R. Ganger

HotCarbon 2024

Acceptance Rate: 46%

DéjàVu: KV-cache Streaming for Fast, Fault-tolerant Generative LLM Serving

Fonteini Strati, [Sara McAllister](#), Amar Phanishayee, Jakub Tarnawski, Ana Klimovic

ICML 2024

Acceptance Rate: 27.5%

Towards Understanding the Carbon Impact in End-to-end Sensing Pipelines

Harsh Desai*, [Sara McAllister*](#), Nathan Beckmann, Brandon Lucia (* = co-first author)

HotEthics 2024

CS-JEDI: Required DEI Education, by CS PhD Students, for CS PhD Students

Bailey Flanigan, Ananya Joshi, [Sara McAllister](#), Catalina Vajiac

🏆 SIGCSE 2023

Acceptance Rate: 35%

Kangaroo: Theory and Practice of Caching Billions of Tiny Objects on Flash

[Sara McAllister](#), Benjamin Berg, Julian Tutuncu-Macias, Juncheng Yang, Sathya Gunasekar, Jimmy Lu, Daniel S. Berger, Nathan Beckmann, Gregory R. Ganger

ACM TOS

August 2022

Kangaroo: Caching Billions of Tiny Objects on Flash

[Sara McAllister](#), Benjamin Berg, Julian Tutuncu-Macias, Juncheng Yang, Sathya Gunasekar, Jimmy Lu, Daniel S. Berger, Nathan Beckmann, Gregory R. Ganger

🏆 SOSP 2021

Acceptance Rate: 16%

External-memory Dictionaries in the Affine and PDAM Models

Michael A. Bender, Alex Conway, Martin Farach-Colton, William Jannen, Yizheng Jiao, Rob Johnson, Eric Knorr, [Sara McAllister](#), Nirjhar Mukherjee, Prashant Pandey, Donald E. Porter, Jun Yuan, Yang Zhan

ACM ToPC

September 2021

The CacheLib Caching Engine: Design and Experiences at Scale

Benjamin Berg, Daniel S. Berger, [Sara McAllister](#), Isaac Grosof, Sathya Gunasekar, Jimmy Lu, Michael Uhlar, Jim Carrig, Nathan Beckmann, Mor Harchol-Balter, Gregory R. Ganger

OSDI 2020

Acceptance Rate: 18%

Small Refinements to DAM Can Have Big Consequences for Data-Structure Design

Michael A. Bender, Alexander Conway, Martin Farach-Colton, William Jannen, Yizheng Jiao, Rob Johnson, Eric Knorr, [Sara McAllister](#), Nirjhar Mukherjee, Prashant Pandey, Donald E. Porter, Jun Yuan, Yang Zhan

SPAA 2019

Acceptance Rate: 40%

Invited Talks

Scaling the IO wall with Declarative IO

OSDI – Seattle, WA

14 July 2026

Salesforce – Database Reading Group, San Francisco, CA

4 Dec 2024

PDL Retreat – Bedford, PA

15 Oct 2024

PDL Retreat – Bedford, PA

6 Nov 2023

Taking off the compute-colored glasses: Storage is vital to datacenter sustainability

PDL Summer Talk Series – Pittsburgh, PA

5 June 2025

Stanford University – Stanford, CA

22 Apr 2025

Microsoft Research – Redmond, WA

21 Apr 2025

University of Wisconsin, Madison – Madison, WI

1 Apr 2025

University of Toronto – Toronto, Canada

27 Mar 2025

Cornell Tech – <i>New York City, NY</i>	20 Mar 2025
MIT – <i>Boston, MA</i>	6 Mar 2025
University of Maryland, College Park – <i>College Park, MD</i>	27 Feb 2025
Google – <i>Sunnyvale, CA</i>	25 Feb 2025

A Call for Research on Storage Emissions

PDL Retreat – <i>Bedford, PA</i>	15 Oct 2024
Western Digital – <i>San Jose, CA</i>	5 Sep 2024
HotCarbon – <i>Santa Cruz, CA</i>	9 July 2024

FairyWREN: A Sustainable Cache for Write-Read-Erase Interfaces

OSDI – <i>San Jose, CA</i>	12 July 2024
PDL Retreat – <i>Bedford, PA</i>	7 Nov 2023
PDL Retreat – <i>Pittsburgh, PA</i>	7 Nov 2022

Towards Understanding the Carbon Impact in End-to-end Sensing Pipelines

HotEthics – <i>Co-presented with Harash Desai, San Diego, CA</i>	29 Apr 2024
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Overcoming Write Limitations to achieve Sustainable Flash Caching

AMD – <i>Santa Clara, CA</i>	29 Mar 2024
Salesforce – <i>Database Reading Group, San Francisco, CA</i>	27 Mar 2024
University of California, Berkeley – <i>Berkeley, CA</i>	25 Jan 2024
Stanford – <i>Stanford, CA</i>	24 Jan 2024
University of California, Santa Cruz – <i>Santa Cruz, CA</i>	11 Jan 2024
McGill University – <i>Montreal, Canada</i>	16 Nov 2023
Microsoft – <i>Pittsburgh, PA</i>	2 Nov 2023
MIT – <i>Boston, MA</i>	10 Oct 2023
NDSEG 2021 Fellows Conference – <i>San Antonio, TX</i>	31 July 2023
University of Toronto – <i>Toronto, Canada</i>	20 Mar 2023

CS-JEDI: DEI education by PhD students, for PhD students

McGill – <i>Montreal, Canada</i>	31 Oct 2023
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Caching on Flash: Kangaroo and Beyond

Meta – <i>Menlo Park, CA</i>	11 Mar 2022
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Kangaroo: Caching Billions of Objects on Flash

Microsoft Research – <i>Redmond, WA</i>	22 Nov 2021
SOSP – <i>Remote</i>	27 Oct 2021
Cache@Scale – <i>Industry Remote Caching Meetup hosted by Meta</i>	4 Mar 2021

Building a Stronger, More Just Academic Community Through Mandatory Anti-bias Learning

University of Pittsburgh Diversity Forum – <i>Co-presented w/ Bailey Flanigan and Catalina Vajiac, Pittsburgh, PA</i>	28 July 2021
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Teaching

University of Wisconsin-Madison

Advanced Operating Systems (CS 736)	Fall 2026
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Carnegie Mellon University

Storage Systems (15-746/18-746)	TA, Fall 2023
Parallel Computer Architecture and Programming (15-418/618)	TA, Spring 2022
Diversity, Equity, and Inclusion in Computer Science and Society (15-996)	Co-Creator and TA, Spring 2021

Harvey Mudd College

Programming Languages (CS131)	Grader and Tutor, Spring 2019
Introduction to Computer Systems (CS105)	Grader and Tutor, Fall 2018
Introduction to Computer Systems (CS105)	Grader and Tutor, Spring 2018
Data Structures and Programming Development (CS70)	Grader and Tutor, Fall 2017
Principles of Computer Science (CS60)	Grader and Tutor, Spring 2017

Guest Lecturer

Computer Systems – Sustainable Computing (CMU 15-213/513)	Fall 2024
Storage Systems – Overcoming Flash's Write Limitations to Achieve Sustainable Caching (CMU 15/18-746)	Fall 2023
Graduate Computer Architecture – Sustainable Computing (CMU 15-740)	Fall 2023
Computer Systems – Kangaroo: Caching Billions of Tiny Objects on Flash (CMU 18-213/613)	Fall 2022
Data Center Computing – Kangaroo Discussion (CMU 18-847C)	Spring 2022
CS-JEDI – Panel on Allyship (CMU 15-996)	Spring 2022
Computer Systems – Kangaroo: Caching Billions of Tiny Objects on Flash (CMU 18-213)	Fall 2021
Storage Systems – Kangaroo: Caching Billions of Tiny Objects on Flash (CMU 18-746)	Fall 2021

Mentoring

Anwesha Saha. Google PhD Student Researcher	Summer 2026
Leo Han. Google PhD Student Researcher	Summer 2026
Theo Gregersen. CMU CS PhD student	Fall 2024 - Present
Tim Kim. CMU CS PhD student	Spring 2024 - Present
Sanjith Athlur. CMU CS PhD student	Spring 2024 - Fall 2025
Sophia (Qingyang) Cao. CMU CS undergraduate student, After degree: Salesforce	Fall 2023 - Spring 2025
Yiwei Chen. CMU ECE masters student	Fall 2024 - Fall 2025
Lucy Wang. CMU ECE undergraduate student	Spring 2024 - Spring 2025
Sarvesh Tandon. CMU ECE masters student	Fall 2023 - Spring 2025
Suhas Thalanki. CMU computational data science masters capstone	Spring 2024 - Fall 2024
Sriya Ravi. CMU computational data science masters capstone	Spring 2024 - Fall 2024
Yu Liu. CMU computational data science masters capstone	Spring 2024 - Fall 2024
Sherry (Yucong) Wang. CMU ECE undergraduate student, After degree: Salesforce	Fall 2022 - Spring 2024
Akshath Karanam. CMU ECE masters student, After degree: Salesforce	Fall 2022
Priyal Suneja. University of Washington CS PhD student	Fall 2021 - Summer 2022
Julian Tutuncu-Macias. CMU CS undergraduate student, After degree: Goldman Sachs	Fall 2019 - Spring 2021
Sheng Xu. CMU CS masters student, After degree: Amazon Web Services	Spring 2020

Service**Program Committee Member**

Symposium on Operating Systems Principles (SOSP)	2026
USENIX Symposium on Operating Systems Design and Implementation (OSDI)	2026

Artifact Evaluation Chair

USENIX Symposium on Operating Systems Design and Implementation (OSDI)	2026
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PhD Admissions

University of Wisconsin-Madison, Computer Science Department	2026
Carnegie Mellon University, Computer Science Department	2022

Student Research Competition (SRC) Committee Member

Symposium on Operating Systems Principles (SOSP)	2026
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Journal Reviewer

Transactions on Storage (TOS)	2024
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External Reviewer

ACM SIGMETRICS	2025
USENIX Annual Technical Conference (ATC)	2024

Faculty Hiring Committee

Carnegie Mellon University, Computer Science Department	2024
Harvey Mudd College, Computer Science Department	2019

Community Outreach

TechNights volunteer – <i>CS program for middle school girls</i>	2019-2020
AP CS Remote Talk at Eagan High School – <i>On CS career opportunities</i>	2021
Science Bus Volunteer and Treasurer – <i>Teaching 4/5th graders at under-resourced schools science lessons</i>	2015 - 2018
STEAM:coders Site Coordinator and Instructor – <i>CS program for students from disadvantaged communities</i>	2016